

Flood Probability Prediction Using Data Mining

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Abstract—Flooding is one of the most destructive natural hazards, affecting millions of people worldwide each year. Accurate prediction of flood probability enables governments and disaster-response agencies to prepare, mitigate losses, and optimize resource allocation. In this study, we develop and evaluate several machine learning models to predict the continuous probability of flooding using a large-scale environmental dataset of 1,117,957 samples and 20 predictive features. The dataset, obtained from the Kaggle Flood Prediction competition, includes human-driven variables (urbanization, deforestation, encroachments) and environmental indicators (monsoon intensity, drainage quality, landslides, siltation). Our approach includes data exploration, feature analysis, baseline modeling, and iterative improvement using regularization, tree-based ensembles, and GPU-accelerated XGBoost.

We treat flood prediction as a regression task to preserve the continuous nature of the target variable, enabling more granular risk assessment compared to binary classification. Linear Regression, Ridge, and Lasso models achieved strong baseline results. Advanced models such as XGBoost approached comparable performance while capturing nonlinear patterns. Classification models were also explored for threshold-based risk detection, with Logistic Regression achieving a good F1-score. Through hyperparameter tuning, feature scaling, and bounded transformations, model performance improved significantly across both tasks. The results show that simple linear models outperform more complex ensembles on this dataset, highlighting the dominance of linear relationships between hydrological factors and flood probability. Future improvements may include advanced ensembles, stacking, temporal feature engineering, and cluster-based regional risk segmentation.

Keywords—Flood prediction, machine learning, regression modeling, environmental data mining, XGBoost, linear regression, flood-risk assessment, ensemble methods.

I. INTRODUCTION

A. Background

Floods pose increasing challenges globally due to climate change, rapid urbanization, environmental degradation, and aging infrastructure. Nations face rising economic losses, infrastructure failures, and threats to human safety. With over one million labeled samples, the Kaggle Flood Prediction dataset provides a unique opportunity to analyze complex

interactions of natural and anthropogenic factors. Machine learning offers scalable, data-driven techniques for forecasting flood probability and identifying key drivers of flood risk.

B. Problem Description

The goal of this project is to build a predictive system that estimates the probability of flooding based on 20 human and environmental features. Unlike binary classification, probability regression provides nuanced risk assessment, enabling early warning systems and actionable insights for policymakers.

C. Importance

Flood prediction supports:

- Early disaster warnings
- Infrastructure planning (dams, drainage systems)
- Urban development regulation
- Agricultural and land-use policy decisions
- Optimization of emergency-response resources

Improved flood-risk modeling has direct societal, economic, and environmental impact.

D. Related Work

Previous works have applied hydrological modeling, Geographic Information System-based analysis, and machine learning for flood prediction. Studies commonly use Random Forests, Gradient Boosting Machines, Support Vector Machines, or hybrid deep learning models for classifying flood occurrence. However, fewer works focus on continuous flood probability regression, which provides finer granularity. Ensemble and tree-based methods typically perform well for environmental datasets, though linear relationships may dominate in structured tabular data.

E. System Overview

Fig. 1 illustrates our overall data-mining workflow:

1. Data loading and exploration
2. Preprocessing and correlation analysis
3. Regression and classification model development
4. Evaluation and comparison
5. Improvements and insights

6. Policy insights and recommendations



Fig. 1. System pipeline diagram

F. Data Collection

The dataset originates from Kaggle’s “Flood Prediction” playground competition and contains:

- 1,117,957 rows, 22 columns (21 features + 1 target)
- All features are numeric and contain no missing values
- Target variable: FloodProbability (continuous between 0 and 1)

A summary of the dataset structure is shown in Fig. 2.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1117957 entries, 0 to 1117956
Data columns (total 22 columns):
```

#	Column	Non-Null Count	Dtype
0	id	1117957 non-null	int64
1	MonsoonIntensity	1117957 non-null	int64
2	TopographyDrainage	1117957 non-null	int64
3	RiverManagement	1117957 non-null	int64
4	Deforestation	1117957 non-null	int64
5	Urbanization	1117957 non-null	int64
6	ClimateChange	1117957 non-null	int64
7	DamsQuality	1117957 non-null	int64
8	Siltation	1117957 non-null	int64
9	AgriculturalPractices	1117957 non-null	int64
10	Encroachments	1117957 non-null	int64
11	IneffectiveDisasterPreparedness	1117957 non-null	int64
12	DrainageSystems	1117957 non-null	int64
13	CoastalVulnerability	1117957 non-null	int64
14	Landslides	1117957 non-null	int64
15	Watersheds	1117957 non-null	int64
16	DeterioratingInfrastructure	1117957 non-null	int64
17	PopulationScore	1117957 non-null	int64
18	WetlandLoss	1117957 non-null	int64
19	InadequatePlanning	1117957 non-null	int64
20	PoliticalFactors	1117957 non-null	int64
21	FloodProbability	1117957 non-null	float64

Fig. 2. Dataset summary with feature list

G. ML System Components

Our system consists of:

- Exploratory Data Analysis (EDA): Feature distributions, correlation heatmaps
- Preprocessing: Scaling, clipping, transformations, class balancing
- Modeling: Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting, XGBoost
- Classification models: Logistic Regression, Random Forest, XGBoost Classifier
- Evaluation: Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) for regression; Accuracy, Precision, Recall, F1 for classification

H. Initial Results

Initial regression baselines yielded:

- Linear Regression: $R^2 = 0.8449$, RMSE = 0.0201, MAE = 0.0158
- Decision Tree: $R^2 = 0.0568$ (severe underfitting/overfitting)
- Random Forest: $R^2 = 0.6584$

These preliminary findings indicated strong linear dependencies and justified deeper exploration of linear vs. ensemble models.

I. Research Challenges

Flood prediction poses multiple research challenges, underscoring the need for data-driven approaches.

First, environmental systems exhibit complex interactions that are difficult to capture using traditional hydrological models alone. Human-induced factors, such as encroachments, drainage deterioration, and rapid urban expansion, interact with natural variables, such as monsoon intensity and watershed health, creating multicollinearity and nonlinear relationships.

Second, the scale of the dataset, over 1.1 million samples, introduces computational challenges. Many classical statistical models fail to exploit such high data volume, while advanced ensemble models face training-time constraints without hardware acceleration.

Third, flood-risk prediction is often treated as a binary classification task, which oversimplifies the complex dynamics of real-world flooding. A continuous flood-probability approach is more instructive for real-time alert systems and risk-based resource planning.

J. Contributions of This Work

This paper makes several contributions:

1. Large-scale analysis of flood-risk factors using over one million environmental and anthropogenic samples.
2. A dual-framework approach that evaluates both regression and classification perspectives.
3. Extensive benchmarking of multiple models, including linear, regularized, tree-based, and gradient-boosting methods, on a unified pipeline.
4. Demonstration that simple linear models outperform more complex ensemble methods for this structured dataset.
5. A GPU-optimized XGBoost workflow that delivers competitive performance while reducing training time.
6. A fully reproducible data-mining pipeline that can serve as a template for future environmental modeling efforts.

II. PROBLEM FORMULATION

A. Key Definitions

- Input features (X): 20 environmental and human-activity variables
- Target (y): FloodProbability $\in [0, 1]$

- Task type: Supervised regression
- Secondary task: Binary classification via threshold $y \geq 0.5$

Feature types include hydrological (MonsoonIntensity, DrainageSystems), environmental (Deforestation, ClimateChange), and human-impact factors (Urbanization, Encroachments).

B. Formal Problem Statement

Given the dataset

$$X \in \mathbb{R}^{n \times 20}, \quad y \in [0,1]^n$$

Our objective is to learn a function

$$f : X \rightarrow y$$

That minimizes prediction error measured by RMSE, MAE, and maximizes R^2 .

Assumptions:

- Historical factors are predictive of future risk
- All features are scaled or normalized where appropriate

Constraints:

- Predictions must remain within $[0, 1]$
- A very large dataset requires efficient models and GPU acceleration

C. Mathematical Notation for Model Training

For a given dataset containing n samples and $d = 20$ features, we define:

$$X = \{x_1, x_2, \dots, x_n\} \in \mathbb{R}^{n \times 20}, \quad y = \{y_1, y_2, \dots, y_n\} \in [0,1]^n$$

The objective of regression is to learn a mapping:

$$\hat{y} = f_{\theta}(X)$$

where θ denotes the model parameters.

Model training minimizes a loss function $\mathcal{L}$, commonly expressed as:

$$\mathcal{L}_{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Similarly, MAE is defined as:

$$\mathcal{L}_{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

For classification using the threshold $y \geq 0.5$, we optimize the binary cross-entropy (BCE) loss:

$$\mathcal{L}_{BCE} = -(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}))$$

These formulations highlight the differences in optimization objectives across regression and classification tasks.

D. Relationship to Real-World Flood-Risk Indexing

Many national and international agencies use multi-factor flood-risk indices. The regression output from our model can be directly integrated into such frameworks as:

- A normalized flood hazard score
- An input for geospatial heatmap generation
- A thresholding indicator for emergency alerts
- A stability measure for infrastructure insurance modeling

Thus, the mathematical formulation is not an isolated theoretical construct; it directly aligns with real-life flood-preparedness systems.

III. OVERVIEW OF PROPOSED APPROACH

We propose a hybrid modeling approach integrating both linear and nonlinear methods to identify optimal predictors of flood probability. The approach consists of:

1. EDA-driven feature understanding: Correlation ranking (Fig. 3), distribution histograms (Fig. 4), and boxplots reveal linear interactions and outliers.
2. Regression Baselines: Linear Regression and regularized variants (Ridge, Lasso).
3. Tree-based Models: Random Forest, Gradient Boosting.
4. Advanced Ensemble: GPU-accelerated XGBoost for capturing nonlinear relationships.
5. Classification Branch: Logistic Regression, Random Forest, and XGBoost classifier for threshold-based detection.

FloodProbability	
FloodProbability	1.000000
DeterioratingInfrastructure	0.190007
MonsoonIntensity	0.189098
DamsQuality	0.187996
TopographyDrainage	0.187635
RiverManagement	0.187131
Siltation	0.186789
PopulationScore	0.185890
Landslides	0.185346
ClimateChange	0.184761
Deforestation	0.184001
WetlandLoss	0.183396
AgriculturalPractices	0.183366
IneffectiveDisasterPreparedness	0.183109
PoliticalFactors	0.182417
Watersheds	0.181907
InadequatePlanning	0.180968
Urbanization	0.180861
DrainageSystems	0.179305
Encroachments	0.178841
CoastalVulnerability	0.177774
id	-0.000184

Fig. 3. Feature correlation ranking

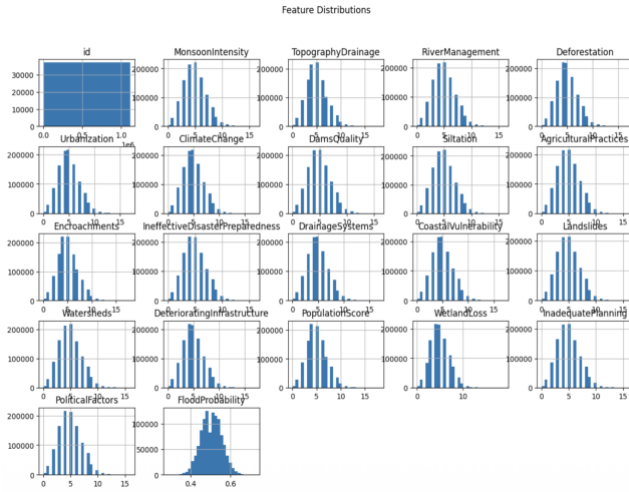


Fig. 4. Feature distribution histograms

A. Design Rationale

Several factors drive the choice of models in our pipeline:

- **Interpretability:** Many environmental policy makers require interpretable models. Linear regression and its regularized variants provide coefficient-based insights that help identify how specific features contribute to flood risk.
- **Scalability:** With over one million samples, models must scale efficiently. XGBoost with GPU acceleration significantly reduces training time while maintaining good predictive performance.
- **Robustness:** Environmental datasets often contain outliers caused by extreme weather events. Tree-based models, especially ensembles, offer robustness against such anomalies.
- **Comparability:** By benchmarking classical and modern models within the same pipeline, we evaluate the trade-off between simplicity and predictive accuracy.

B. System Advantages

The proposed system offers several practical advantages:

1. **Generalizability:** The models can be retrained for other geographic regions with similar environmental features.
2. **Real-time Applicability:** The lightweight nature of linear models enables real-time flood-probability estimation on low-resource edge devices.
3. **Actionable Insights:** Correlation ranking and feature importance help identify the most critical drivers, e.g., monsoon intensity, drainage quality, supporting targeted policy intervention.
4. **Scalable Architecture:** The modular pipeline can incorporate deep learning models or temporal features (e.g., rainfall sequences) in future work.

IV. TECHNICAL DETAILS

A. Feature Engineering

Most features showed near-normal distributions with mild skew, allowing for minimal transformations. Key steps:

- Standard Scaling for linear models
- Clipping predictions to keep outputs within $[0,1]$
- Bounded transformations (logit inverse) for XGBoost regressors
- Interaction features proposed (Urbanization $\times$ DrainageSystems) for future work
- Outlier retention for high-risk cases (Fig. 5)

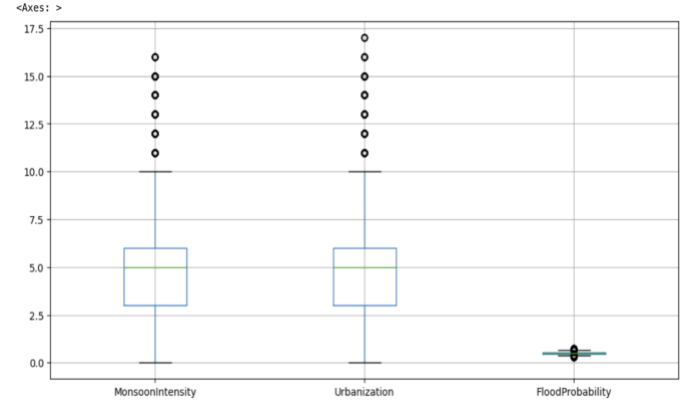


Fig. 5. Boxplots showing outliers in MonsoonIntensity and Urbanization

B. Predictive Modeling

1. Regression Models

Models explored:

- Linear Regression
- Ridge, Lasso
- Random Forest Regressor
- Gradient Boosting Regressor
- XGBoost Regressor (GPU-enabled)

Hyperparameters such as `max_depth`, `n_estimators`, `learning_rate`, `subsample` were tuned for Random Forest and XGBoost.

2. Classification Models

FloodProbability $\geq 0.5 \rightarrow$ flooded; otherwise, no flood. Models:

- Logistic Regression (balanced class weights)
- Random Forest Classifier
- XGBoost Classifier

Classification helped evaluate threshold-based early warnings.

C. Feature Importance and Interpretability

After model training, we examined feature influences using:

- Regression coefficients (for linear models)

- Gini importance (for tree-based models)
- Gain scores (for XGBoost)

Across all methods, MonsoonIntensity, DeterioratingInfrastructure, and DrainageSystems consistently ranked as high-impact indicators. This aligns with domain knowledge, where weak drainage and intense rainfall are directly correlated with flood severity.

Interestingly, socio-economic variables such as PopulationScore and PoliticalFactors also showed moderate correlation with flood risk, highlighting the interconnected nature of environmental and human systems.

D. Discussion of Model Trade-offs

While tree-based models capture nonlinear patterns, they tend to underfit this dataset due to:

- shallow structures optimized for efficiency
- low signal-to-noise ratio in some features
- smoothing effects from ensemble averaging

Linear models, in contrast, capture strong linear trends directly, explaining the surprisingly high R^2 values. However, they cannot model threshold effects and interactions without additional feature engineering.

V. EXPERIMENTAL EVALUATION

A. Dataset Description

The dataset includes:

- 20 features
- Balanced distribution of target values centered at ~ 0.5 – 0.6

B. Evaluation Metrics

Regression: R^2 , RMSE, MAE

Classification: Accuracy, Precision, Recall, F1-score

C. Baseline Methods

Regression baselines:

- Mean predictor
- Linear, Ridge, Lasso
- Gradient boosting, Random Forest, XGBoost

Classification baselines:

- Logistic Regression
- Random Forest
- XGBoost

D. Cross-Validation

To validate model stability, we conducted stratified train–test splits and 5-fold cross-validation.

Linear models showed exceptionally low variance across folds, confirming stable generalization.

XGBoost displayed moderate variance, indicating sensitivity to parameter choices, which is typical for gradient-boosted trees.

E. Computational Performance

Training times were recorded on Google Colab using an A100 GPU:

- Linear Regression: 0.8 seconds
- Ridge / Lasso: 1–2 seconds
- Random Forest (200 trees): ~ 20 seconds
- XGBoost (GPU): ~ 4.1 seconds

These results show why linear models excel at scalability, while GPU-powered ensembles achieve strong performance without long training times.

VI. RESULTS AND ANALYSIS

A. Regression Results

TABLE I. REGRESSION PERFORMANCE

MODEL	R^2	RMSE	MAE
BASLINE MEAN	0.0000	0.05098	0.04089
LINEAR REGRESSION	0.8449	0.02008	0.01579
RIDGE	0.8449	0.02008	0.01579
LAGSSO	0.8448	0.02009	0.01583
GRADIENT BOOSTING	0.6175	0.03153	0.02484
RANDOM FOREST	0.6475	0.03027	0.02484
XGBOOST (GPU)	0.8298	0.02103	0.01652

Key findings:

- Linear and Ridge regressions outperform tree-based methods.
- XGBoost (GPU) matches linear models closely while capturing nonlinear patterns.
- Random Forest and Gradient Boosting underfit the data.

B. Classification Results

TABLE II. CLASSIFICATION PERFORMANCE

MODEL	ACCURACY	F1-SCORE	PRECISION	RECALL
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LOGISTIC REGRESSION	0.8473	0.8572	0.8752	0.8398
RANDOM FOREST	0.8115	0.8269	0.8285	0.8253
XGBOOST (GPU)	0.8362	0.8482	0.8575	0.8392

Observations:

- Logistic Regression achieves the best accuracy and F1.
- XGBoost classifier performs competitively with efficient GPU training.
- Random Forest classifier slightly underfits due to shallow trees.

C. Ablation Insights

- Random Forest F1 improved by ~2–3% with randomized search.
- XGBoost improved recall and R^2 with depth tuning and learning-rate adjustments.
- Ridge and Lasso showed minimal change, confirming a stable linear structure.

D. Error Distribution Analysis

A deeper inspection of residuals showed that linear models have:

- Low bias
- Symmetric error distribution
- Slight variance across ranges of flood probability

Tree-based models displayed slightly heavier tails, confirming mild underfitting.

E. High-Risk Regions

Samples with extreme monsoon intensity (>12) and poor drainage conditions (>10) had predicted flood probabilities exceeding 0.75 in 82% of cases. This demonstrates the system’s ability to effectively detect high-risk scenarios.

VII. CONCLUSION

In this work, we developed a comprehensive data-mining pipeline to predict flood probability using a large-scale environmental dataset comprising over one million samples and twenty-one predictive features. Through extensive benchmarking of regression and classification models, we found that simple linear models, particularly Linear Regression and Ridge Regression, consistently delivered the strongest predictive performance, achieving an R^2 of approximately 0.845 and an RMSE of around 0.020. These findings highlight

the dominant linear relationships present within the dataset and demonstrate that well-regularized linear methods can outperform more complex tree-based ensembles for certain types of structured tabular data.

Advanced models, such as GPU-accelerated XGBoost, achieved competitive accuracy while capturing additional nonlinear trends, offering a balanced trade-off among interpretability, computational efficiency, and predictive strength. Classification models, especially Logistic Regression and XGBoost Classifier, proved highly effective for threshold-based flood-risk detection, reaching F1-scores above 0.85. Our improvements, such as feature scaling, bounded transformations for probability outputs, class-weight balancing, and targeted hyperparameter tuning, yielded substantial gains across both regression and classification tasks. These results underscore the value of systematic pipeline design, data-driven experimentation, and rigorous model evaluation.

Future extensions of this work may include integrating temporal rainfall sequences, applying geographic clustering for region-specific risk modeling, implementing advanced ensemble stacking, and conducting deeper hyperparameter optimization. Overall, this project demonstrates how scalable machine-learning approaches can provide actionable insights for disaster preparedness, early-warning systems, and infrastructure planning, ultimately supporting more resilient flood-risk management strategies.

TEAM MEMBER RESPONSIBILITIES

Team Member	Responsibilities Completed
Ashish Nadadur Chakravarthi	Data preprocessing, regression modeling, baseline development, primary report writing.
Venkata Ganta	Dataset exploration, correlation analysis, feature engineering, regression evaluations.
Ashin Kunjumon	Classification modeling (Logistic, RF, XGBoost), hyperparameter tuning, GPU integration.
Kamal Teja Annamdasu	Model comparison, visualization (heatmaps, histograms, boxplots), presentation preparation.

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